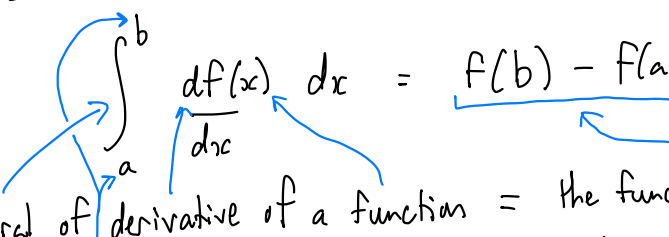


The Fundamental Theorem of Geometric Calculus Jan 3rd 2026

Something I have been wanting to understand better for a while.
Prompted by Joseph sharing insights about analytic complex functions.
Following Doran & Lasenby Chapter 6.

In a nutshell

The fundamental theorem of calculus (1-d version)



The diagram shows the equation $\int_a^b \frac{df(x)}{dx} dx = f(b) - f(a)$. Blue arrows indicate the mapping from the components of the equation to a descriptive sentence. An arrow from the integral symbol points to 'integral of'. An arrow from the derivative $\frac{df(x)}{dx}$ points to 'derivative of a function'. An arrow from the differential dx points to '(over some domain)'. An arrow from the entire left side of the equation points to 'integral of derivative of a function ='. An arrow from the difference $f(b) - f(a)$ points to 'the function integrated over the domain's boundary'.

$$\int_a^b \frac{df(x)}{dx} dx = f(b) - f(a)$$

integral of derivative of a function = the function integrated over
(over some domain) the domain's boundary.

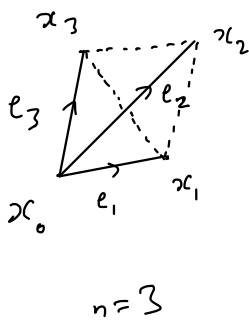
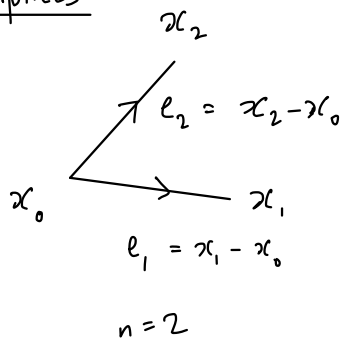
Many versions of this statement exist in higher dimensions:

1. Green's Theorem
2. The Divergence Theorem
3. Stokes' Theorem
4. Cauchy's Integral Formula
5. Generalized Stokes.

That 1, 2 and 3 are special cases of 5 is well-known. That 4 is also, is less so. FTG-C is the Geometric Algebra version of 5.

Simplices

(2)



x_i are position vectors.

Barycentric words

Directed vol. element: $\Delta X = \frac{1}{n!} e_1 \wedge \dots \wedge e_n$

Point in simplex: $x = x_0 + \sum_{i=1}^n \lambda^i e_i$ where

$\left. \begin{array}{l} 0 \leq \lambda^i \leq 1 \\ \sum_{i=1}^n \lambda^i \leq 1 \end{array} \right\} (2i)$

$$\int dX = \int_0^1 \int_0^{1-\lambda^1} \dots \int_0^{1-\sum_{i=2}^n \lambda^i} \underbrace{e_1 \wedge \dots \wedge e_n}_J d\lambda^1 \dots d\lambda^n$$

$$= \int_0^1 \int_0^{1-\lambda^1} \dots \int_0^{1-A} d\lambda^2 \dots d\lambda^n \cdot J$$

$A = \sum_{i=3}^n \lambda^i$

note:
 λ^i is not λ to the power of i , but the i 'th λ .

$$\left[-\frac{1}{2} (1-\lambda^2-A)^2 \right]_0^{1-A} = -\frac{1}{2} (0 - (1-A)^2) = \frac{1}{2} (1-A)^2$$

$$= \int_0^1 \int_0^{1-\lambda^1} \dots \int_0^{1-B} \frac{1}{2} (1-\lambda^3-B)^2 d\lambda^3 \dots d\lambda^n \cdot J \quad B = \sum_{i=4}^n \lambda^i$$

$$\xrightarrow{\quad} \frac{1}{3!} (1-B)^3$$

$$= \int_0^1 \frac{1}{(n-1)!} (1-\lambda^n)^{n-1} d\lambda^n \cdot J = \frac{1}{n!} J = \Delta X \quad (2ii)$$

Notation for k -simplex: $(x)_{(k)} = (x_0, x_1, \dots, x_k)$

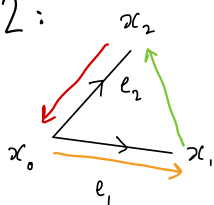
(3)

ie just the set of (position vectors of)
 $k+1$ points defining the simplex.

Boundary of the simplex:

$$\partial(x)_{(k)} = \sum_{i=0}^k (-1)^i \underbrace{(x_0, \dots, \overset{\vee}{x_i}, \dots, x_k)}_{(k-1)\text{-simplex formed by removing } x_i} \quad (3i)$$

Eg $k=2$:



$$\partial(x)_{(2)} = \underbrace{(x_1, x_2)}_{\substack{\uparrow \\ \text{remove } x_0 \\ (-1)'}} - \underbrace{(x_0, x_2)}_{\substack{\uparrow \\ \text{remove } x_1 \\ (-1)'}} + \underbrace{(x_0, x_1)}_{\substack{\uparrow \\ \text{remove } x_2 \\ (-1)'}}$$

Boundary of a boundary vanishes:

$$\partial \partial(x)_{(k)} = 0.$$

$$\text{Eg: } \partial \partial(x)_{(2)} = (x_2) - (x_1) - [(x_2) - (x_0)] + (x_1) - (x_0) = 0$$

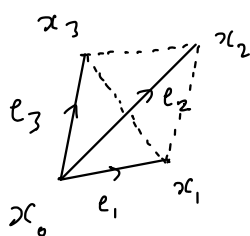
(Directed) volume:

$$\Delta x \equiv \Delta(x)_{(k)} = \frac{1}{k!} e_1 \wedge \dots \wedge e_k = \frac{1}{k!} (x_1 - x_0) \wedge (x_2 - x_0) \wedge \dots \wedge (x_k - x_0)$$

Directed volume of a boundary vanishes:

$$\Delta(\partial(x)_{(k)}) = 0.$$

To verify, distribute Δ over the sum in the boundary expression. For example, $k=3$:



$$\partial(x)_{(3)} =$$

(4)

$$(x_1, x_2, x_3) - (x_0, x_2, x_3) + (x_0, x_1, x_3) - (x_0, x_1, x_2)$$

$$\Delta \partial(x)_{(3)} = \frac{1}{2!} \left[\underbrace{(x_2 - x_1)(x_3 - x_1)}_{\text{blue}} - \underbrace{(x_2 - x_0) \wedge (x_3 - x_0)}_{\text{green}} + \underbrace{(x_1 - x_0) \wedge (x_3 - x_0)}_{\text{red}} - \underbrace{(x_1 - x_0) \wedge (x_2 - x_0)}_{\text{orange}} \right]$$

$$= \frac{1}{2!} \left[\begin{aligned} & \cancel{x_2 \wedge x_3} - \cancel{x_2 \wedge x_1} - \cancel{x_1 \wedge x_3} \quad | \\ & - \cancel{x_2 \wedge x_3} + \cancel{x_2 \wedge x_0} + \cancel{x_0 \wedge x_3} \quad | \\ & + \cancel{x_1 \wedge x_3} - \cancel{x_1 \wedge x_0} - \cancel{x_0 \wedge x_3} \quad | \\ & - \cancel{x_1 \wedge x_2} + \cancel{x_1 \wedge x_0} + \cancel{x_0 \wedge x_2} \quad | \end{aligned} \right] = 0$$

Note that the directed integral over the boundary of a simplex vanishes:

$$\oint_{\partial(x)_{(k)}} dS = \sum_{k=0}^k (-1)^i \int_{(\check{x}_i)_{(k-1)}} dX = \Delta \partial(x)_{(k)} = 0 \quad (4i)$$

Distribute integral over the sum in the boundary expression

In a simplex chain, the directed areas of common faces cancel, so can fill a space and only integrate over the space boundary.

Multivector Fields on Simplices

(5)

(barycentric)

The $\{e_i\}$ are linearly independent, so define a frame, albeit not orthonormal. Can therefore linearly interpolate a multivector field $F(x)$ as $f(x) = F_0 + \sum_{i=1}^n \lambda^i (F_i - F_0)$

where $F_i = F(x_i)$. So, can write integral of F over simplex boundary as:

$$\oint_{\partial(x)_{(n)}} f(x) dS = \sum_{i=1}^n (F_i - F_0) \oint_{\partial(x)_{(n)}} \lambda^i dS$$

ΓF_0 term vanishes
because constant
- see (4i)

Now, want to get something like $\int_{(x)_{(n)}} \nabla F dx$ on the RHS,

and $F_i - F_0$ is already in the ballpark of a derivative. In fact,

$$F_i - F_0 = \frac{\partial f}{\partial \lambda^i}$$

Given $x = x_0 + \sum_{i=1}^n \lambda^i e_i$ from page (2), we have $\lambda^i = e^i \cdot (x - x_0)$ where $\{e^i\}$ is the frame reciprocal to $\{e_i\}$, so that $e^i \cdot e_j = \delta^i_j$.

So, we are dealing with something of the form

$$\oint_{\partial(x)_{(n)}} b \cdot x dS \quad (5i)$$

where b is e^i and the x_0 term vanishes (see (4i))

Because the integral over a simplex boundary is the antisymmetrized $\textcircled{6}$ sum over simplices one dimension lower, it pays to consider integrals over simplices themselves as on page $\textcircled{2}$. We already have that

$$\int_{(x)(k)} dX = \Delta X = \frac{1}{k!} e_1 \wedge \dots \wedge e_k, \text{ to repeat (2.ii).}$$

(can show similarly that $\int_{(x)(k)} \lambda^i dX = \frac{1}{k+1} \Delta X \quad \forall i :$

$$\int_0^1 \int_0^{1-\lambda^k} \dots \int_0^{1-A^i} \lambda^i \frac{(1-\lambda^i-A^i)^{i-1}}{(i-1)!} d\lambda^i \dots d\lambda^k \cdot J \quad A^i = \sum_{j=i+1}^k \lambda^j$$

$\int u'v = uv - \int uv'$

$$\frac{1}{(i-1)!} \left(\left[\frac{(1-\lambda^i-A^i)^i}{i} \lambda^i \right]_0^{1-A^i} - \int_0^{1-A^i} \frac{(1-\lambda^i-A^i)^i}{i} d\lambda^i \right) = \frac{(1+A^i)^{i+1}}{(i+1)!}$$

The λ^i in the i 'th integral increments the power and factorial by an extra 1, which gives at the end

$$\int_{(x)(k)} \lambda^i dX = \frac{1}{(k+1)!} J = \frac{1}{k+1} \Delta X \quad (6i)$$

From (5i), focus on $b \cdot x$. In barycentric coordinates (2i), (7)

$$b \cdot x = b \cdot x_0 + \sum_{i=1}^k b_i \lambda^i \quad \text{where } b_i = b \cdot e_i$$

$$\text{So } \int_{(x)_{(k)}} b \cdot x \, dX = b \cdot x_0 \Delta X + \sum_{i=1}^k b_i \int_{(x)_{(k)}} \lambda^i \, dX \quad (6i)$$

$$= \left(b \cdot x_0 + \frac{1}{k+1} \sum_{i=1}^k b_i \right) \Delta X$$

→ geometric center of simplex, $\bar{x}$, because:

$$\sum_{i=1}^k b_i = \sum_{i=1}^k b \cdot (x_i - x_0) = \sum_{i=1}^k b \cdot x_i - k b \cdot x_0$$

$$\begin{aligned} \text{So } b \cdot x_0 + \frac{1}{k+1} \sum_{i=1}^k b_i &= \frac{1}{k+1} \left((k+1) b \cdot x_0 - \cancel{k b \cdot x_0} + \sum_{i=1}^k b \cdot x_i \right) \\ &= \frac{1}{k+1} \sum_{i=0}^k b \cdot x_i \quad \text{note} \end{aligned}$$

So we have that:

$$\int_{(x)_{(k)}} b \cdot x \, dX = b \cdot \bar{x} \Delta X \quad (7i)$$

However, (5i) integrates over $\partial(x)_{(k)}$ (the boundary), not $(x)_{(k)}$ (the simplex), so let's address that next.

From (3i),

(8)

$$\oint_{\partial(\mathcal{X})_{(k)}} b \cdot x \, dS = \sum_{i=0}^k (-1)^i \int_{(\check{x}_i)_{(k-1)}} b \cdot x \, dX \quad \text{where } (\check{x}_i)_{(k-1)} \text{ is the } k-1 \text{ simplex formed by omitting } x_i \text{ from } (\mathcal{X})_{(k)}.$$

$$= \sum_{i=0}^k (-1)^i \frac{1}{k} b \cdot (x_0 + \dots + \check{x}_i + \dots + x_k) \Delta(\check{x}_i)_{(k-1)} \text{ by (7i)}$$

$$= \frac{b \cdot X}{k} \underbrace{\sum_{i=0}^k (-1)^i \Delta(\check{x}_i)_{(k-1)}}_{=0 \text{ by (4i)}} + \frac{1}{k} \sum_{i=0}^k (-1)^i b \cdot x_i \Delta(\check{x}_i)_{(k-1)} \quad \text{where } X = \sum_{i=0}^k x_i$$

$$= \frac{1}{k} \sum_{i=0}^k (-1)^i b \cdot (\underbrace{x_i - x_0}_{\text{constant}}) \Delta(\check{x}_i)_{(k-1)} \quad \text{so can add in by (4i)}$$

$$= \frac{1}{k} \sum_{i=0}^k (-1)^i b \cdot \underbrace{e_i}_{\text{using the identity:}} \frac{1}{(k-1)!} e_1 \wedge \dots \wedge \check{e}_i \wedge \dots \wedge e_k$$

$$= \frac{1}{k!} b \cdot (e_1 \wedge \dots \wedge e_k) = \sum_{i=1}^k (-1)^{i+1} b \cdot a_i (a_1 \wedge \dots \wedge \check{a}_i \wedge \dots \wedge a_r)$$

$$\text{So, } \oint_{\partial(\mathcal{X})_{(k)}} b \cdot x \, dS = b \cdot \Delta X \quad (8i)$$

This is the key result that gives us the FTGC applied to page (5).

On page ⑤ we had:

$$\begin{aligned}
 \oint_{\partial(x)_{(n)}} f(x) dS &= \sum_{i=1}^n (F_i - F_0) \oint_{\partial(x)_{(n)}} \lambda^i dS \\
 &= \sum_{i=1}^n \frac{\partial f}{\partial \lambda^i} \oint_{\partial(x)_{(n)}} e^i \cdot (x - x_0) dS \quad \text{by (4i)} \\
 &= \sum_{i=1}^n \frac{\partial f}{\partial \lambda^i} e_i \cdot \Delta X \quad \text{by (8i)} \\
 &= \dot{f} \cdot \dot{\nabla} \cdot \Delta X \quad \text{where } \dot{\nabla} = \sum_i e^i \partial_i \text{ in barycentric coords.}
 \end{aligned}$$

Now, all of this is for one simplex. By filling a space V with a simplex chain $\{\Delta(x)_{(n)}^a\}$ and taking the limit of vanishing simplex size, we get

$$\oint_{\partial V} F dS = \int_V \dot{F} \dot{\nabla} dX \quad (9i) \quad \left(\lim_a \sum \dot{F} \dot{\nabla} \cdot \Delta X^a \right)$$

note: inner product removed because outer product contributes zero because grade > dims of V .

Rework the above with multivector field to right of measure:

$$\oint_{\partial V} dS G = \int_V \dot{\nabla} dX \dot{G} \quad (9ii)$$

Can generalize further by defining a linear function $L(A_{n-1}; x)$ (10) consuming a grade $n-1$ blade and returning a multivector of any grades. Linear interpolation:

$$L(A; x) = L(A; x_0) + \sum_{i=1}^n \lambda^i (L(A; x_i) - L(A; x_0))$$

Linearity means commutation with all sums/integrals, and we get:

$$\oint_{\partial V} L(dS) = \int_V \dot{L}(\dot{V} dX) \quad (10i) \quad (\text{position-dep suppressed})$$

Applications

The Divergence Theorem follows from (10i) with:

$$(n\text{-dim}) \quad L(A) = \langle J A I^{-1} \rangle \quad \text{for a vector field } J.$$

$$\langle J dS I^{-1} \rangle$$

grades $1 \quad n-1 \quad n$

$$= \langle J n | dS | \rangle$$

$$= n \cdot J |dS|$$

$$\text{where } n |dS| = dS I^{-1}$$

$$\langle \dot{J} \dot{V} dX I^{-1} \rangle$$

grades $1 \quad 1 \quad n \quad n$

scalar

$$|dX| = I^{-1} dX$$

scalar

$$\nabla \cdot J$$

(anticommuting term vanishes as grade 2)

$$\therefore \oint_{\partial V} n \cdot J |dS| = \int_V \nabla \cdot J |dX|$$

Green's Theorem follows from (9ii) with G being the 2-d vector field $J = P e_1 + Q e_2$ ($e_i \cdot e_j = \delta_{ij}$)

(11)

$$\oint_{\partial V} dS J = \int_V \underbrace{\nabla \cdot dX}_{\text{grades } 2, 1} J = - \int_V \nabla J dX$$

vector J anticommutes with 2-d pseudoscalar $dX = I dx dy$

$$\begin{aligned} \nabla J &= (\partial_x e_1 + \partial_y e_2) (P e_1 + Q e_2) \\ &= \partial_x P + \partial_y Q + (\partial_x Q - \partial_y P) e_1 e_2 \end{aligned}$$

LHS is a scalar. Scalar part of $\nabla J I = -(\partial_x Q - \partial_y P)$

$$\therefore \oint P dx + Q dy = \int (\partial_x Q - \partial_y P) dx dy.$$

Stokes' Theorem arises when $L(a) = J \cdot a$ for vectors J and a .

So (10i) becomes

$$\oint_{\partial V} J \cdot dL = \int_V J \cdot (\underbrace{\nabla}_{1} \cdot \underbrace{dX}_{2})$$

$\wedge$ term is grade 3 but LHS is scalar

$$= - \int_V (\nabla \wedge J) \cdot dX$$

because $a \cdot (b \cdot B)$

$$= (a \wedge b) \cdot B$$

for vectors a, b and bivector B .

For Cauchy's integral formula, need the notion of an analytic complex function, or more generally a monogenic multivector field.

Let $\varphi = ue_1 + ve_2$ and $\psi = e, \varphi = u + vI$ be a 2-d vector field and the corresponding complex function, which is analytic when the Cauchy-Riemann conditions are met:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \quad \text{or} \quad u_x = v_y \quad \text{and} \quad u_y = -v_x.$$

This means the map defined by the complex function is locally conformal. Small changes are well-approximated by the Jacobian

$$\begin{pmatrix} u_x & v_x \\ u_y & v_y \end{pmatrix} = \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \quad \text{given } \begin{matrix} u_x = a \\ v_x = b \end{matrix} \quad \text{and C.R.}$$

$$= w \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \quad \text{for } \begin{matrix} w^2 = a^2 + b^2 \\ \tan \phi = \frac{b}{a} \end{matrix}$$

uniform
dilation

rotation

"magnitwist" only - no shearing.

Squares can get bigger/smaller and change angle,
but do not become parallelograms.

This was Joseph's insight.

Monogenic functions satisfy $\nabla \Psi = 0$: (13)

$$\begin{aligned}\nabla \Psi &= (d_x e_1 + d_y e_2)(u + v e_1 e_2) \\ &= (\underline{u_x - v_y}) e_1 + (\underline{u_y + v_x}) e_2 = 0\end{aligned}$$

CR

The vector derivative of the corresponding vector field shows the magnitude explicitly:

$$\begin{aligned}\nabla \Psi &= \nabla \cdot \Psi + \nabla \wedge \Psi \\ &= (d_x e_1 + d_y e_2)(u e_1 + v e_2) \\ &= (u_x + v_y) + (v_x - u_y) e_1 e_2 = 2a + 2bI = 2we^{I\theta}.\end{aligned}$$

The FTGC in the form of (9ii) with $G = \Psi$ is

$$\oint_{\partial V} \frac{\partial r}{\partial \lambda} \Psi d\lambda = \int_V \nabla \Psi dX$$

$\nearrow$
 curve param

Ψ commutes with dX
 because even-grade.

$r = x e_1 + y e_2$

Map to complex plane by premultiplying by e_1 :

$$e_1 \oint_{\partial V} \frac{\partial r}{\partial \lambda} \Psi d\lambda = \oint_{\partial V} \frac{\partial z}{\partial \lambda} \Psi d\lambda = \oint \Psi dz = \int \underbrace{e_1 \nabla \Psi}_{=0 \text{ if } \Psi \text{ monogenic}} dX \quad (13i)$$

$\therefore$ have Cauchy's Theorem: $\oint \Psi dz = 0$ for analytic Ψ .

Cauchy's integral formula (CIF) is :

(14)

$$f(a) = \frac{1}{2\pi i} \oint \frac{f(z)}{z-a} dz \quad \text{for analytic } f(z).$$

Now: $\frac{1}{z-a} = \frac{(z-a)^{\dagger}}{|z-a|^2} = \frac{(r-b)e_1}{(r-b)^2}$ as $z^{\dagger} = (e, r)^{\dagger} = re_1$,
 $a^{\dagger} = be_1$,

and $\frac{r-b}{(r-b)^2} = \nabla \ln |r-b|$. We also know that $\ln |r-b|$ is

the Green function for ∇^2 in 2-d: $\nabla^2 \ln |r-b| = 2\pi \delta(r-b)$

so $\nabla \frac{r-b}{(r-b)^2} = \nabla^2 \ln |r-b| = 2\pi \delta(r-b)$ existence of Green fn
↕
 ∇ invertible.

Now let $\psi = \frac{f(z)}{z-a}$ in (13i):

$$\begin{aligned} \oint \frac{f(z)}{z-a} dz &= e_1 \int \nabla \left(\frac{r-b}{(r-b)^2} e_1 g(r) \right) dX \\ &= e_1 \int \left(\underbrace{2\pi \delta(r-b)}_{\downarrow} e_1 g(r) + \cancel{\nabla f(r)} \frac{r-b}{(r-b)^2} e_1 \right) \underbrace{I |dX|}_{\substack{\text{0 for analytic } f. \\ \text{If } f \text{ not analytic, now we} \\ \text{have generalized Cauchy's} \\ \text{result!}}} \end{aligned}$$

$= 2\pi I f(a)$

$\frac{1}{z-a}$ is the Green function for the vector derivative in 2-d, and C.I.F. propagates $f(z)$ off the boundary to any point a .